

Software Engineering Lab

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Section: E

Scenario: #43 Feature Flag & Dynamic Config Manager system

Chosen architecture: We chose Microservices Architecture for the Feature Flag & Dynamic Config Manager system.

Reasons:

- The system handles two very distinct workloads: frequent, high-throughput real-time flag checks and infrequent administrative modifications. A microservices approach isolates the flag evaluation engine from the flag & rollout management component. This ensures that administrative updates or complex audit logging tasks never block or interfere with active evaluation paths
- The client SDK synchronization and evaluation traffic experience heavy, fluctuating loads depending on application activity. Microservices allow us to independently scale the evaluation and sync components horizontally without having to scale the entire application

Security: By segmenting the system into discrete components, we enforce strict access control boundaries (which is NFR-002). Administrative operations, like creating flags, adjusting rollout percentages and viewing audit logs, are strictly isolated under the management and observability components. This makes sure that unauthorized users or external calls cannot compromise critical production configuration

Performance benefit: The architecture allows the flag evaluation engine to operate with extreme efficiency. Since it is decoupled from heavy database writes and auditing overhead, the engine easily satisfies the strict performance constraint of executing evaluations in under 15 ms (for NFR-001)

Component diagram:

